

| Question | Marking Guidance                                                                                                                                                                                       | Mark | Comments                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------|
| 08.1     | 1. (Short) single strand of DNA;<br>2. <u>Bases</u> complementary (with DNA/allele/gene);                                                                                                              | 2    |                                                                             |
| 08.2     | 1. <u>Restriction</u> endonuclease/enzyme;<br>2. (Cuts DNA at specific) base sequence<br><b>OR</b><br>(Breaks) phosphodiester <u>bonds</u><br><b>OR</b><br>(Cuts DNA) at recognition/restriction site; | 2    | 2. Accept palindromic sequence.                                             |
| 08.3     | (So DNA) probe binds/attaches/anneals;                                                                                                                                                                 | 1    |                                                                             |
| 08.4     | 1. (Lane 1 has DNA fragments) of known sizes/lengths;<br>2. Compare (position of viral fragment/s);                                                                                                    | 2    |                                                                             |
| 08.5     | 3, 4, 5 with these numbers in any sequence;                                                                                                                                                            | 1    | All three numbers required.<br><br>Reject if more than three numbers given. |

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